# Case Study: Customer Churn Analysis

## Understanding Policyholder Churn and Retention Opportunities in a Life Insurance Portfolio

### Background

A life insurance company manages a diverse portfolio of policyholders across a range of insurance products and customer profiles. Maintaining long-term relationships with policyholders is important to the sustainability of the business. However, not all policyholders remain active throughout their relationship with the company. Some customers eventually discontinue their insurance relationship with the company. This phenomenon, commonly referred to as *customer churn*, can affect recurring premium revenue and increase the need to continuously acquire new customers. Although *customer churn* is an important business concern, identifying the underlying patterns is not straightforward. Customers may differ in terms of demographic characteristics, insurance products, premium levels, payment preferences, policy tenure, and information related to claims. Patterns observed within one customer group may not necessarily apply to others.

The company has collected historical data on its customers and policies, including customer characteristics and information about their insurance relationships with the company. Management wants to use these data to better understand *churn* patterns across the portfolio and generate insights that can support decision-making. Management also needs to understand which parts of the portfolio require further attention, as not every *churn* pattern identified carries the same business implications.

### Business Question

How can the company use data to better understand and respond to the issue of customer churn in a targeted manner?

### Participant Task

Participants will act as a Data Analytics Team supporting the company's management. In this final stage, the task is not simply to build a predictive model, but to investigate the issue of customer churn from multiple analytical perspectives. Participants are expected to identify patterns or findings that have business relevance, investigate them using appropriate analytical approaches, and translate their findings into actionable business recommendations.

There is no single prescribed analytical methodology for this challenge. The four challenges below provide direction for the participants' investigation but are not intended to limit the scope of their analysis. Participants may develop additional analytical questions and select methods they consider appropriate based on the data and business context provided.

## Analytical Considerations

Participants may use predictive models or statistical methods as part of their analysis where appropriate. However, model performance alone is not the primary objective of this challenge. The value of a solution lies in how effectively participants use the data to investigate the problem, support their conclusions with appropriate evidence, and formulate actionable recommendations.

Participants who use predictive models are expected to apply appropriate evaluation and validation approaches to assess model performance and reduce the risk of overfitting. For other quantitative methods, participants are expected to apply appropriate robustness checks or other relevant evaluation approaches to assess the reliability of their analytical findings. Participants should explain the rationale behind their chosen approach and its limitations.

The analysis should also take into account the limitations of the available data. In particular, observed patterns or relationships should be interpreted with caution, and associations identified in the data should not be treated as evidence of causal relationships without appropriate justification.

## Challenges

The four challenges below serve as guiding questions for the participants' investigation. They are intended to help keep the analysis focused on the business problem while leaving room for participants to pursue additional analytical questions where relevant.

## Challenge 1: Understanding the Churn Landscape

What are the key patterns and characteristics of *churn* across the company's portfolio?

Participants are expected to explore the data to develop an overall understanding of the *churn* landscape and identify patterns, differences, or characteristics that are relevant to the business problem.

Participants are free to determine which aspects of the data are most relevant to investigate and are not limited to any particular variables or analytical approach.

## Challenge 2: From Patterns to Business Priorities

Which parts of the portfolio warrant greater attention in addressing *churn*, and where are there meaningful opportunities to improve retention?

Not every pattern observed in the data has the same business relevance. Participants are expected to assess their findings from a business perspective and identify customer groups, conditions, or areas within the portfolio that are particularly relevant to the opportunity to improve retention.

The prioritization of areas for further attention should be supported by evidence from the data and grounded in the business context of the problem. Participants are not required to use any specific method or analytical approach to make this assessment.

## Challenge 3: Investigating Factors Associated with Churn

Which findings provide the strongest evidence of an association with *churn*, and how strong is the evidence supporting them?

Participants are expected to investigate further the areas or findings identified as relevant in the earlier stages of the analysis in order to better understand the factors or patterns associated with *churn*.

The analysis should not only identify differences or associations, but also assess the strength, consistency, and reliability of the evidence supporting the findings. Participants may use statistical methods, predictive models, or other analytical approaches that are appropriate for the questions being investigated.

Participants should also consider the limitations of the available data and possible alternative explanations for the relationships observed. Observed associations should not be interpreted as evidence of causal relationships without appropriate justification.

## Challenge 4: Developing Retention Strategies

Based on the findings from the investigation, what retention strategies or interventions should the company consider, and which policyholders or conditions should be the primary focus?

Participants are expected to translate their analytical findings into recommendations that are relevant to the business problem.

Recommendations should be supported by findings from the analysis and clearly explain the problem, customer group, or conditions targeted by the proposed action. Participants should also explain how the insights identified through the analysis can be translated into practical actions for the company.